SATRAJIT SUJIT GHOSH

Curriculum Vitae

McGovern Institute for Brain Research
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Department of Otolaryngology – Head and Neck Surgery
Massachusetts Eye and Ear
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Degrees

PhD, Cognitive and Neural Systems, Boston University, 2005, Prof. Frank Guenther
B.S. (Honors), Computer Science, National University of Singapore, 1997, Prof. Lonce L. Wyse

Employment

Senior Research Scientist, McGovern Institute for Brain Research, MIT, 2025 – Current
Director of the Open Data in Neuroscience Initiative, McGovern Institute for Brain Research, MIT, 2023 – Current
Assistant Professor, Department of Otolaryngology, Harvard Medical School, 2014 – Current
Principal Research Scientist, McGovern Institute for Brain Research, MIT, 2015 – 2025
Research Scientist, McGovern Institute for Brain Research, MIT, 2011 – 2014
Research Scientist, Research Laboratory of Electronics, MIT, 2007 – 2011
Postdoctoral Associate, Research Laboratory of Electronics, MIT, 2004 – 2007, Dr. Joseph S. Perkell
Software Engineer, Kent Ridge Digital Labs, Singapore, 1997-1998

External Positions held

Massachusetts Eye and Ear, 2014 – Current, Research Associate
Speech and Hearing Biosciences and Technology (SHBT), Division of Medical Sciences, Harvard Medical School, 2011 – Current, Member of the Faculty
Program in Neuroscience, Division of Medical Sciences, Harvard Medical School, 2019 – Current, Member of the Faculty
Speech and Hearing Biosciences and Technology (SHBT), Health Science and Technology (HST), 2008 – 2011, Member of the Faculty
Standards for Datasharing Taskforce, International Neuroinformatics Coordinating Facilities, 2010 – 2016
Executive board, TankThink Labs, LLC, 2011 – 2015
Department of Cognitive and Neural Systems, Boston University, 2005-2010, Research Fellow

Honors

Winner - Predictive Analytics Competition for Depression, University of Muenster, 2018
Phase I winner for the Open Science Prize competition, NIH, HHMI, Wellcome Trust, 2016
Educational stipend, International Society for Magnetic Resonance in Medicine, 2008
Graduate Teaching Fellow Award, Boston University, 2000
Presidential University Graduate Fellowship, Boston University, 1998

MIT UROP Students supervised

Alkhairy, Samiya, Fall, 2009, Spring 2010

Zhang, Mark, Spring 2012
Ung, William, Spring 2012
Smith, Ashley, Spring 2015
Biswas, Jyotishka, Spring 2016
Suh, Michelle, Spring 2016
Taylor, Tilly, Spring 2016
Jackson, Blake, Spring 2016
Batmunkh, Zulsar, Spring 2016
Wu, David, Fall 2016
Wu, Kathy, Spring 2017
Shumaev, Alexander, Fall 2018
Moreno, Felipe, Fall 2018
Megha Vemuri, Fall, Spring, Summer 2022
Nicholas F Gustafson, Fall, 2022
Sabeen Lohawala, Spring, 2023
Veronika Moroz, Spring, 2025

MEng/MSc Thesis Supervised
Alice Bizeul, EPFL, 2020
Gasser Elbanna, EPFL, 2023
Agathe Tournant, ETH, 2024
Sabeen Lohawala, MIT, 2024
Miles Silva, MIT, 2025
Yassine Jamaa, EPFL, 2025
Alina Weinberger, TUM, 2025
Jordan Wilke, MIT, In progress
Bruke Wossenseged, MIT, In progress

Ph.D. Students Supervised
Ciccarelli, Gregory, EECS, MIT, Characterization of Phone Rate as a Vocal Biomarker of Depression, 2017 (Current: Amazon, Inc.)
Sitek, Kevin, SHBT, Harvard University, Investigating the human subcortical auditory pathway with MRI, 2019 (Current: Research Assistant Professor, Northwestern University)
Low, Daniel, SHBT, Harvard University, Speech and text psychometrics: Identifying suicide risk factors with large language models and acoustic networks, 2024 (Current: Research Scientist, Child Mind Institute)
Mentch, Jefffrey, SHBT, Harvard University, In progress
Burdinski, Debbie, BBS MD/PhD, Harvard University, In progress
Rahul Brito, SHBT, Harvard University, In progress

Postdoctoral Researchers Supervised
Ghosh, Debanjan, 2018 – 2019 (Current: Educational Testing Service, Princeton, NJ)
Padhy, Smruti, 2016 – 2018 (Current: Research Associate, Texas Advanced Supercomputing Center, TX)
Jarecka, Dorota, 2016 – 2017 (Current: Research scientist, MIT)
Rajaei, Hoda, 2019 – 2022 (Current: Machine Learning Scientist, Beyond Limits, CA)
Kleinberger, Rebecca, 2020 – 2021 (Current: Assistant Professor, Northeastern University)
Rana, Aakanksha, 2020 – 2022 (Current: Senior Scientist, Imaging AI, Johnson & Johnson)
Das, Dhritiman, 2020 - 2022

King, Maedbh, 2022 - 2025
Catania, Fabio, 2022 -
Chan, Yibei, 2023 –
Chhetri, Tek Raj, 2024 -

Teaching experience
6.541/SHBT.204, Speech Communication, Spring 2009, 2011- 2016
6.551/SHBT.200, Acoustics of Speech and Hearing, Fall 2007- 2015
9.S912, Quantitative Methods and Computational Models in Neuroscience, Fall 2015
HST.583, fMRI Data Acquisition and Analysis, Fall 2015, 2017, 2019
HST.714/SHBT.200/9.016, Introduction to Sound, Speech, and Hearing, Fall 2016 – Fall 2022
SHBT.205, Speech and Hearing: From Neuroscience to Perception, Spring 2024 -

Service

Internal service:
Committee on Research Computing and Data, Office of Research Computing and Data, MIT, 2022 - Current
Admissions committee, Speech and Hearing Biosciences and Technology Program (HST), 2010 – Current
Curriculum committee, Speech and Hearing Biosciences and Technology Program (HST), 2009 – Current
Director, Openmind Neuroscience High Performance Computing Resource, 2014 - 2023
Chair, BCS Faculty Committee on Computational Infrastructure, 2019 - 2022

External service:

Scientific Advisory Board
INCF, Council for Training, Science and Infrastructure (CTSI), Chair, 2024 -
NIH Healthy Brain and Child Development Study (https://hbcdstudy.org), 2021 -
OpenScope Project Allen Institute for Brain Science (https://openscope.ai), 2021-
NWB – Neurodata Without Borders (https://nwb.org), 2020 -
CONP – Canadian Open Neuroscience Platform (https://conp.ca), 2018 - 2019
SINDS – Neurohackademy Training Program (https://neurohackademy.org), 2017 - 2022

Editorial board
Aperture Neuro, Handling editor, 2020 -
Frontiers in Brain Imaging Methods, 2012 – 2022
Frontiers in Neuroinformatics, 2016 – 2022
Frontiers in Human Neuroscience, 2015 – 2017

Member
NIH Study Section, Neurological, Mental and Behavioral Health (NMBH), Chair, 2025 –
NIH Study Section, Neurological, Mental and Behavioral Health (NMBH), Standing Member, 2023 – 2025
Organization for Human Brain Mapping (OHBM) - Best Practices Committee, 2020 - 2024
Alzheimer's Drug Discovery Foundation - Diagnostics Accelerator Speech Consortium, 2020

Ad hoc grant reviewer
EU Horizon (2023)
National Institute of Health, 2017 – 2022 (NOIT, BRAIN Initiative, EITN)
National Science Foundation, 2008, 2010, 2013
National Medical Research Council, Singapore, 2007, 2009, 2011-2012

Department of Defense, 2011
Simons Foundation, 201
Israel Science Foundation, 2015, 2019

Ad hoc editorial reviewer

Biological Psychiatry
Brain
Brain and Language
Brain Structure and Function
Cerebral Cortex
Current Biology
Elife
European Journal of Neuroscience
Frontiers in Computational Neuroscience
Frontiers in Systems Neuroscience
Frontiers in Neuroinformatics
Human Brain Mapping
Journal of the Acoustical Society of America
Journal of Machine Language Research
Journal of Neuroscience
Journal of Speech, Language and Hearing Research
Magnetic Resonance in Medicine
Nature Methods
Nature Translational Psychiatry
NeuroImage
Neuroinformatics
Neuron
PLOS One
PLOS Computational Biology

Editorial board, Special Research Topic, Python in Neuroscience I / II, Frontiers in Neuroscience
Nipype teaching workshops, Edinburgh 2011, Magdeburg 2012, Boston 2017
Speaker, Educational workshop, Organization for Human Brain Mapping, 2013, 2018
Organizer, HBM Hackathon, Organization for Human Brain Mapping, Seattle, 2013
Local organizing committee, 4th Biennial Conference on Resting State Connectivity, Boston, 2014

Technological and Other Scientific Innovations

DANDI	https://dandiarchive.org An opensource infrastructure as a service supported by the US NIH BRAIN Initiative to house and disseminate neurophysiology data for the international neuroscience community. https://github.com/dandi
Methods and Apparatus for Reducing Stuttering	US Patent No.: 11727949 – Issued August 2023 This is based on dissertation work of former postdoctoral associate Rebecca Kleinberger. It involves an apparatus and a software to enable realtime vocal modification as an assistive technology for individuals who stutter.

An application (Mumble Melody) is available on the Apple Store for worldwide use.

Assessing Disorders Through Speech And A Computational Model	U.S. Patent No.: 10127929 – Issued November 2018 1. * Williamson JR, Quatieri TF, Helfer BS, Perricone J, **Ghosh SS**, Ciccarelli G, Mehta DD. (2015) Segment-dependent dynamics in predicting Parkinson's disease. In Sixteenth Annual Conference of the International Speech Communication Association. 2. * Ciccarelli G, Quatieri TF, **Ghosh SS** (2016) Neurophysiological Vocal Source Modeling for Biomarkers of Disease. In Seventeenth Annual Conference of the International Speech Communication Association. The goal of this effort is to supplement the VoiceUp platform with augmented algorithms for tracking mental health state using computational models. The model itself is based on my doctoral thesis work and I guided the team to using that framework. This came out of the MIT MINT funded collaboration with MIT Lincoln Laboratory.
Nipype: Brain imaging analysis framework 2008-	Gorgolewski K, Burns CD, Madison C, Clark D, Halchenko YO, Waskom ML, Ghosh SS. (2011). Nipype: a flexible, lightweight and extensible neuroimaging data processing framework in Python. Front. Neuroimform. 5:13. https://github.com/nipy/nipype Nipype provides an open source Python library for constructing scalable, reusable, and efficient dataflows for biomedical research. Nipype provides a standard Python interface to 750+ tools and algorithms from more than 25 neuroimaging software packages written in C++, MATLAB, Java, and Python. Nipype dataflows can be executed in various HPC (high-performance computing), commercial Cloud, and local environments. Nipype forms a base software layer for some of the most popular neuroimaging workflows in use today (fMRIPrep, Mindboggle, C-PAC and others). Over 200+ individuals have contributed to the code base and the software is used in 88+ countries.
MURFI: a realtime MR biofeedback software 2007	Hinds, O., **Ghosh, S.**, Thompson, T.W., Yoo, J.J., Whitfield-Gabrieli, S., Triantafyllou, C., Gabrieli, J.D. (2011) Computing moment-to-moment BOLD activation for real-time neurofeedback. Neuroimage. 54(1):361-8. PMID: 20682350. https://github.com/gablab/murfi2/ This opensource software framework allows biofeedback of activation based on the BOLD signal. I created the testing and validation framework for the software and contributed to its design and implementation. We are now using this software for ongoing projects in the treatment of schizophrenia and in the development of new paradigms. Development of this was supported by the McGovern Institute MINT program.

	This is now in use in two clinical trials at Northeastern University and at University of Minnesota, Minneapolis.
Realtime vocal modification software 2005	Cai, S., **Ghosh, S.**, Guenther, F., Perkell, J. (2011). Focal manipulations of formant trajectories reveal a role of auditory feedback in the online control of both within-syllable and between-syllable speech timing. J Neurosci 31: 45. 16483-16490. PMID: 22072698. https://github.com/shanqing-cai/audapter_matlab https://github.com/shanqing-cai/audapter_mex This opensource framework allows modifying vocal characteristics in realtime. I established the initial framework and guided Marc Boucek and Shanqing Cai in extending the framework to perform new paradigms.
Noise suppression for MRI patient microphone input 2004	Two provisional patents were applied for but not pursued after expiry. 2007 Online noise suppression software for Magnetic Resonance Imaging 2007 Bidirectional noise suppressing communication setup for Magnetic Resonance Imaging https://arxiv.org/abs/1207.5827 The goal of this software was to provide a mechanism to suppress MR noise. This is still being used in research projects at MIT.
Carotid artery diameter estimation from ultrasound images 1999	Current usage status is unknown. I built the graphical interface for the software to provide a semi-automated method for artery diameter estimation that reduced human intervention significantly and validated it against manual measurements.
FlexEffex: Interactive sound effects and music 1997	I contributed to the development of the FlexEffex architecture and rewrote the internal sound effects plugin api and hardware libraries. The software was subsequently sold to a company, MindMaker Inc.

Publications (* = PhD students are authors; + = co-equal authorship role)

1. Guenther, F.H., Nieto-Castanon, A., Tourville, J.A. and **Ghosh, S.S.** (2001) The effects of categorization training on auditory perception and cortical representations. Proceedings of the Speech Recognition as Pattern Classification (SPRAAC) Workshop, Nijmegen, The Netherlands.
2. Guenther, F.H. and **Ghosh, S.S.** (2003) A model of cortical and cerebellar function in speech. Proceedings of the XVth International Congress of Phonetic Sciences (pp. 169-173). Barcelona, Spain: 15th ICPhS Organizing Committee.
3. Guenther, F.H., **Ghosh, S.S.** and Nieto-Castanon, A. (2003) A neural model of speech production. Proceedings of the 6th International Seminar on Speech Production. Sydney, Australia
4. Nieto-Castanon, A., **Ghosh, S.S.**, Tourville, J.A., Guenther, F.H. (2003) Region of interest based analysis of functional imaging data. Neuroimage. 19(4):1303-16. PMID: 12948689.
5. Guenther, F.H., Nieto-Castanon, A., Ghosh, S.S., Tourville, J.A. (2004) Representation of sound categories in auditory cortical maps. J Speech Lang Hear Res. 47(1):46-57. PMID: 15072527.

6. Max, L., Guenther, F.H., Gracco, V.L., **Ghosh, S.S.** and Wallace, M.E. (2004) Unstable or insufficiently activated internal models and feedback-biased motor control as sources of dysfluency: A theoretical model of stuttering. Contemporary Issues in Communication Science and Disorders. 31.
7. Klein, A., Mensh, B., **Ghosh, S.**, Tourville, J., Hirsch, J. (2005) Mindboggle: automated brain labeling with multiple atlases. BMC Med Imaging. 5:7. PMCID: PMC1283974.
8. Guenther, F.H., **Ghosh, S.S.**, Tourville, J.A. (2006) Neural modeling and imaging of the cortical interactions underlying syllable production. Brain Lang. 96(3):280-301. PMCID: PMC1473986.
9. Guenther, F.H., **Ghosh, S.S.**, Nieto-Castanon, A. and Tourville, J.A. (2006) A neural model of speech production. In: J. Harrington & M. Tabain (eds.), Speech Production: Models, Phonetic Processes, and Techniques. London: Psychology Press.
10. Tiede, M., Shattuck-Hufnagel, S., Johnson, B., **Ghosh, S.**, Matthies, M., Zandipour, M. and Perkell, J. (2007) Gestural phasing in /kt/ sequences contrasting within and cross word contexts. Proceedings of the XVIth International Congress of Phonetic Sciences. Saarbrücken, Germany.
11. **Ghosh, S.S.**, Tourville, J.A., Guenther, F.H. (2008) A neuroimaging study of premotor lateralization and cerebellar involvement in the production of phonemes and syllables. J Speech Lang Hear Res. 51(5):1183-202. PMCID: PMC2652040.
12. Cai, S, Boucek, M, **Ghosh, S.S.**, Guenther, F.H., Perkell, J.S. (2008) A System for Online Dynamic Perturbation of Formant Trajectories and Results from Perturbations of the Mandarin Triphthong /iau/. International Seminar in Speech Production, Strassbourg, France.
13. Balci, S.K., Sabuncu, M.R., Yoo, J., **Ghosh, S.S.**, Whitfield-Gabrieli, S., Gabrieli, J.D., Golland, P. (2008) Prediction of Successful Memory Encoding from fMRI Data. Med Image Comput Comput Assist Interv. 2008(11):97-104. PMCID: PMC2855196.
14. Perkell, J.S., Lane, H., **Ghosh, S.S.**, Matthies, M.L., Tiede, M., Guenther, F., Ménard, L. (2008) Mechanisms of Vowel Production: Auditory Goals and Speaker Acuity. International Seminar in Speech Production, Strassbourg, France.
15. Klein, A., **Ghosh, S.S.**, Avants, B., Yeo, B.T., Fischl, B., Ardekani, B., Gee, J.C., Mann, J.J., Parsey, R.V. (2010) Evaluation of volume-based and surface-based brain image registration methods. Neuroimage. 51(1):214-20. PMCID: PMC2862732.
16. Cai, S., **Ghosh, S.S.**, Guenther, F.H., Perkell, J.S. (2010) Adaptive auditory feedback control of the production of formant trajectories in the Mandarin triphthong /iau/ and its pattern of generalization. J Acoust Soc Am. 128(4):2033-48. PMCID: PMC2981117.
17. **Ghosh, S.S.**, Kakunoori, S., Augustinack, J., Nieto-Castanon, A., Kovelman, I., Gaab, N., Christodoulou, J.A., Triantafyllou, C., Gabrieli, J.D., Fischl, B. (2010) Evaluating the validity of volume-based and surface-based brain image registration for developmental cognitive neuroscience studies in children 4 to 11 years of age. Neuroimage. 53(1):85-93. PMCID: PMC2914629.
18. **Ghosh, S.S.**, Matthies, M.L., Maas, E., Hanson, A., Tiede, M., Ménard, L., Guenther, F.H., Lane, H., Perkell, J.S. (2010) An investigation of the relation between sibilant production and somatosensory and auditory acuity. J Acoust Soc Am. 128(5):3079-87. PMCID: PMC3003728.
19. Golfinopoulos, E., Tourville, J.A., Bohland, J.W., **Ghosh, S.S.**, Nieto-Castanon, A., Guenther, F.H. (2011) fMRI investigation of unexpected somatosensory feedback perturbation during speech. Neuroimage. 55(3):1324-38. PMCID: PMC3065208
20. Silver, A.L., Nimkin, K., Ashland, J.E., **Ghosh, S.S.**, Van der Kouwe, A.J., Brigger, M.T., Hartnick, C.J. (2011) Cine magnetic resonance imaging with simultaneous audio to evaluate pediatric velopharyngeal insufficiency. Arch Otolaryngol Head Neck Surg. 137(3):258-63.
21. Brunner, J., **Ghosh, S.**, Hoole, P., Matthies, M., Tiede, M., Perkell, J. (2011) The influence of auditory acuity on acoustic variability and the use of motor equivalence during adaptation to a perturbation. J Speech Lang Hear Res. 54(3):727-39. PMID: 20966388.

22. Cai, S., **Ghosh, S.**, Guenther, F., Perkell, J. (2011). Focal manipulations of formant trajectories reveal a role of auditory feedback in the online control of both within-syllable and between-syllable speech timing. J Neurosci 31: 45. 16483-16490. PMID: 22072698.
23. Hinds, O., **Ghosh, S.**, Thompson, T.W., Yoo, J.J., Whitfield-Gabrieli, S., Triantafyllou, C., Gabrieli, J.D. (2011) Computing moment-to-moment BOLD activation for real-time neurofeedback. Neuroimage. 54(1):361-8. PMID: 20682350.
24. Gorgolewski, K., Burns, C.D., Madison, C., Clark, D., Halchenko, Y.O., Waskom, M.L., **Ghosh, S.S.** (2011). Nipype: a flexible, lightweight and extensible neuroimaging data processing framework in Python. *Front. Neuroinform.* **5**:13.
25. Perrachione, T.K., Del Tufo, S.N., **Ghosh, S.S.**, Gabrieli, J.D.E. (2011) "Phonetic variability in speech perception and the phonological deficit in dyslexia." 17th Meeting of the International Congress of Phonetic Sciences, (Hong Kong, August 2011).
26. Poline, J., Breeze, J.L., **Ghosh, S.S.**, Gorgolewski, K., Halchenko, Y.O., Hanke, M., Haslegrove, C., Helmer, K.G., Marcus, D.S., Poldrack, R.A., Schwartz, Y., Ashburner, J. and Kennedy, D.N. (2012). Data sharing in neuroimaging research. *Front. Neuroinform.* **6**:9.
27. **Ghosh, S.S.**, Klein, A., Avants, B. and Millman, K.J. (2012). Learning from open source software projects to improve scientific review. *Front. Comput. Neurosci.* **6:**18
28. Cai, S., Beal, D.S., **Ghosh, S.S**., Tiede, M.K., Guenther, F.H., Perkell, J.S. (2012) Weak responses to auditory feedback perturbation during articulation in persons who stutter: Evidence for abnormal auditory-motor transformation. PLoS One.
29. Doehrmann, O.[+], **Ghosh, S.S**.[+], Polli, F.P., Reynolds, G., Horn, F., Keshavan, A., Whitfield-Gabrieli, S., Hofmann, S.G., Pollack, M., Gabrieli, J.D. (2013) Predicting treatment response in social anxiety disorder from functional magnetic resonance imaging. JAMA Psychiatry.
30. Hinds, O., Thompson, T., **Ghosh, S.S.**, Yoo, J., Whitfield-Gabrieli, S., Triantafyllou, C., Gabrieli, J. (2013) Roles of Default-Mode Network and Supplementary Motor Area in Human Vigilance Performance: Evidence from Real-Time fMRI. Journal of Neurophysiology.
31. Tustison NJ, Johnson HJ, Rohlfing T, Klein A, **Ghosh SS**, Ibanez L and Avants B (2013). Instrumentation bias in the use and evaluation of scientific software: Recommendations for reproducible practices in the computational sciences. Front. Neurosci. 7:162.
32. **Ghosh, S.S.**, Keshavan, A., Langs, G (2013). Predicting Treatment Response from Resting State fMRI Data: Comparison of Parcellation Approaches. 3rd International Workshop on Pattern Recognition in NeuroImaging (Philadelphia, June 2013).
33. Perrachione, T.K. and **Ghosh, S.S.** (2013). Optimized design and analysis of sparse-sampling fMRI experiments. Front. Neurosci. 7:55. doi: 10.3389/fnins.2013.00055
34. Cai, S., Beal, D.S., **Ghosh, S.S.**, Guenther, F.H., Perkell, J.S. (2014) Impaired timing adjustments in response to time-varying auditory perturbation during connected speech production in persons who stutter. Brain and Language.
35. Cai, S., Tourville, J.A., Beal, D.S., Perkell, J.S., Guenther, F.H. and **Ghosh, S.S.** (2014). Diffusion Imaging of Cerebral White Matter in Persons Who Stutter: Evidence for Network-Level Anomalies. Front. Hum. Neurosci. 8:54
36. Christodoulou JA, Del Tufo SN, Lymberis J, Saxler PK, **Ghosh SS**, Triantafyllou C, Whitfield-Gabrieli S, Gabrieli JD. (2014). Brain bases of reading fluency in typical reading and impaired fluency in dyslexia. PLoS One. 9(7):e100552. doi: 10.1371/journal.pone.0100552. eCollection 2014.
37. Stoeckel, L.E., Garrison, K.A., **Ghosh, S.S.**, Wighton, P., Hanlon, C.A., Gilman, J.M., Greer, S., Turk-Browne, N.B., deBettencourt, M.T., Scheinost, D., Craddock, C., Thompson, T., Calderon, V., Bauer, C.C., George, M., Breiter, H.C., Whitfield-Gabrieli, S., Gabrieli, J.D., LaConte, S.M., Hirshberg, L., Brewer, J.A., Hampson, M., Van Der Kouwe, A., Mackey, S., Evins, A.E. (2014).

Optimizing real time fMRI neurofeedback for therapeutic discovery and development, NeuroImage: Clinical

38. Gabrieli, J.D.E., **Ghosh, S.S.**, Whitfield-Gabrieli, S. (2015). Prediction as a Humanitarian and Pragmatic Contribution from Human Cognitive Neuroscience. Neuron.
39. Gorgolewski KJ, Varoquaux G, Rivera G, Schwartz Y, Sochat VV, **Ghosh SS**, Maumet C, Nichols TE, Poline JB, Yarkoni T, Margulies DS, Poldrack RA (2015). NeuroVault.org: A repository for sharing unthresholded statistical maps, parcellations, and atlases of the human brain. Neuroimage.
40. Gorgolewski KJ, Varoquaux G, Rivera G, Schwarz Y, **Ghosh SS**, Maumet C, Sochat VV, Nichols TE, Poldrack RA, Poline JB, Yarkoni T, Margulies DS. (2015). NeuroVault.org: a web-based repository for collecting and sharing unthresholded statistical maps of the human brain. Front Neuroinform. 10;9:8.
41. Langs G, Golland P, **Ghosh SS.** (2015) Predicting Activation Across Individuals with Resting-State Functional Connectivity Based Multi-Atlas Label Fusion. Med Image Comput Comput Assist Interv. 9350:313-320.
42. * Williamson JR, Quatieri TF, Helfer BS, Perricone J, **Ghosh SS**, Ciccarelli G, Mehta DD. (2015) Segment-dependent dynamics in predicting Parkinson's disease. Interspeech 16.
43. * Sitek KR, Cai S, Beal DS, Perkell JS, Guenther F and **Ghosh SS** (2016). Decreased cerebellar-orbitofrontal connectivity correlates with stuttering severity: Whole-brain functional and structural connectivity associations with persistent developmental stuttering. Front. Hum. Neurosci. 10:190. doi: 10.3389/fnhum.2016.00190
44. * Ciccarelli G, Quatieri TF, **Ghosh SS** (2016) Neurophysiological Vocal Source Modeling for Biomarkers of Disease. Interspeech 17.
45. Whitfield-Gabrieli S, **Ghosh SS**, Nieto-Castanon A, Saygin Z, Doehrmann O, Chai XJ, Reynolds GO, Hofmann SG, Pollack MH, Gabrieli JD. (2016) Brain connectomics predict response to treatment in social anxiety disorder. Mol Psychiatry.
46. Allen GI, Amoroso N, Anghel C, Balagurusamy V, Bare CJ, Beaton D, Bellotti R, Bennett DA, Boehme K, Boutros PC, Caberlotto L, Caloian C, Campbell F, Chaibub Neto E, Chang YC, Chen B, Chen CY, Chien TY, Clark T, Das S, Davatzikos C, Deng J, Dillenberger D, Dobson RJB, Dong Q, Doshi J, Duma D, Errico R, Erus G, Everett E, Fardo DW, Friend SH, Fröhlich H, Gan J, George-Hyslop P, **Ghosh SS**, Glaab E, Green RC, Guan Y, Hong MY, Huang C, Hwang J, Ibrahim J, Inglese P, Jiang Q, Katsumata Y, Kauwe JSK, Klein A, Kong D, Krause R, Lalonde E, Lauria M, Lee E, Lin X, Liu Z, Livingstone J, Logsdon BA, Lovestone S, Lyappan A, Ma M, Malhotra A, Mangravite LM, Maxwell TJ, Merrill E, Nagorski J, Namasivayam A, Narayan M, Naz M, Newhouse SJ, Norman TC, Nurtdinov RN, Oyang YJ, Pawitan Y, Peng S, Peters MA, Piccolo SR, Praveen P, Priami C, Sabelnykova VY, Senger P, Shen X, Simmons A, Sotiras A, Stolovitzky G, Tangaro S, Tateo A, Tung YA, Tustison NJ, Varol E, Vradenburg G, Weiner MW, Xiao G, Xie L, Xie Y, Xu J, Yang H, Zhan X, Zhou Y, Zhu F, Zhu H, Zhu S. (In press) Crowdsourced estimation of cognitive decline and resilience in Alzheimer's disease, Alzheimer's & Dementia, Available online 11 April 2016, ISSN 1552-5260, http://dx.doi.org/10.1016/j.jalz.2016.02.006.
47. Cameron Craddock R, S Margulies D, Bellec P, Nolan Nichols B, Alcauter S, A Barrios F, Burnod Y, J Cannistraci C, Cohen-Adad J, De Leener B, Dery S, Downar J, Dunlop K, R Franco A, Seligman Froehlich C, J Gerber A, **Ghosh SS**, J Grabowski T, Hill S, Sólon Heinsfeld A, Matthew Hutchison R, Kundu P, R Laird A, Liew SL, J Lurie D, G McLaren D, Meneguzzi F, Mennes M, Mesmoudi S, O'Connor D, H Pasaye E, Peltier S, Poline JB, Prasad G, Fraga Pereira R, Quirion PO, Rokem A, S Saad Z, Shi Y, C Strother S, Toro R, Q Uddin L, D Van Horn J, W Van Meter J, C Welsh R, Xu T (2016). Brainhack: a collaborative workshop for the open neuroscience community. Gigascience. 5:16. doi: 10.1186/s13742-016-0121-x. eCollection 2016. PubMed PMID: 27042293; PubMed Central PMCID: PMC4818387.

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126. Chen Y, Jarecka D, Abraham SA, Gau R, Ng E, Low DM, Bevers I, Johnson A, Keshavan A, Klein A, Clucas J, Rosli Z, Hodge SM, Linkersdörfer J, Bartsch H, Das S, Fair D, Kennedy D, **Ghosh SS**. Standardizing Survey Data Collection to Enhance Reproducibility: Development and Comparative Evaluation of the ReproSchema Ecosystem. J Med Internet Res 2025;27:e63343. doi: 10.2196/63343. PMID: 40644691

Invited Presentations

Challenges in Performing FAIR and Reproducible Computation
FAIR in ML, AI Readiness & Reproducibility (FARR) Workshop, Washington DC, 2024

DANDI: Building a collaborative ecosystem for neuroscientific data
Neuroscience Gateway workshop - Society for Neuroscience Annual Meeting, Chicago, IL, 2024

The transformative potential and challenges of open data and technologies in neuroscience
NIH Data Sharing and Reuse Seminar Series 2024

The Role of Infrastructures in Enabling AI for Neuroscience
Symposium: BRAIN Informatics in the AI Era
BRAIN Initiative Conference 2024

Building an Integrated Ecosystem for Brain Behavior Quantification and Synchronization (BBQS)
Specialty Session: Brain Behavior Quantification and Synchronization (BBQS) - Laying the Foundation for a New Data and Knowledge Ecosystem
BRAIN Initiative Conference 2024

Exploring the Speech Chain: Bridging Science and Technology
Boston University Speech, Language, and Hearing Sciences (SLHS) Colloquium Series, 2024

Data availability, access, and transparency
Study Panel on The Science and Ethics of Measuring and Modeling Individual and Group Behavior, American College of Neuropsychopharmacology (ACNP), 2023

An Emerging Ecosystem for Psychopathology Research
Psychiatry Grand Rounds, Motto Endowed Lecture, University Hospitals and Case Western Reserve University, 2023

The transformative potential and challenges of open data and computation in neuroscience
BBQS Sensors Workshop, NIH, 2023

Using translational applications to unpack machine learning models and systemic challenges
Machine Learning in Medicine Seminar Series, Radiology, WCM & Electrical and Computer Engineering, Cornell-Ithaca and Cornell-Tech, 2023

Seeing precision psychiatry through the variability lens of data and technologies
American Psychopathological Association Conference, 2023

Can neuroinformatics infrastructures like DANDI advance scientific discovery?
NeuroDataShare: Exploring and sharing multi-scale neuroscience data, University College London, 2023

Unpacking the Speech Chain: A window of scientific and technological opportunities
Quantitative Life Sciences Seminar Series, McGill University, 2022

Leveraging brain research to change scientific culture, education, and infrastructure
Precision Convergence Webinar Series, McGill University and Pittsburgh Supercomputing Center, 2022

Towards precision psychiatry through diverse sensors and machine learning
Data Science in Clinical Settings Symposium, Fundación INECO, 2021

Into the neuroverse
OHBM Student and Postdoc Special interest group, 2021

Sensors and the Brain
CANDI Lab Shriver Center, University of Massachusetts Medical School, 2021
OSU-CN Yang Webinar series, Oregon State University, 2021

What constitutes a good standard for neuroscience?
INCF Virtual assembly, 2021

The Shifting Dunes of Data and Computation
University of Virginia, Biomedical Data Science Seminar, 2021

Reproducible Workflows and Analysis
ABCD-Repronim course, Florida International University, 2020

Challenging the Invisibility of Mental Illness
American Medical Informatics Association - INCF, 2020

The evolution of machine learning in brain imaging
Frontiers in Brain Imaging Symposium, University of Texas, Southwestern, 2020

What has working with brains, voice, and infrastructure technologies taught us about open science?
NeuroHub Seminar Series, McGill University, 2019

Retooling Psychiatry: How will we get there?
Computational Psychiatry Symposium, University of Iowa, 2019

Brains, Voice, and Technology: A multifaceted approach to mental health
Center for Depression, Anxiety, and Stress Research, McLean Hospital, 2019

Assistive Intelligence for Brain Health
World Medical Innovation Forum, 2019

Modeling Noise and Individual Variation
Organization for Human Brain Mapping, Singapore, 2018

Tools of the trade: From Data to Results in Neuroimaging
Neuroscience Information Framework, Online Webinar, 2018

Speaking one's mind: Vocal biomarkers of mental health

University of Washington, Seattle, August, 2018

A brain cartographer's quandary
Workshop on large-scale trends in cortical organization, Leipzig, Germany, 2017

Speaking one's mind: Vocal biomarkers of mental health
Technology in Psychiatry, Symposium, Boston, USA, 2017

The emerging informatics revolution in neuroscience
Boston Children's Hospital, Boston, USA, 2017
Center for Addiction Medicine, Massachusetts General Hospital, Boston, USA, 2017
Department of Biomedical Informatics, University of Pittsburgh, Pittsburgh, USA, 2017

Variance is the spice of reproducible research
Keynote: Annual Neuroinformatics Congress, Kuala Lampur, Malaysia, 2017

Applications of Machine Learning to Brain Imaging and Psychiatry
Computational Psychiatry Workshop, Satellite of Biological Psychiatry, San Diego, USA, 2017

Predicting Treatment Outcome in Social Anxiety Disorder and Tracking Major Depression and Parkinson State Using Behavioral Information
ACNP 55th Annual Meeting, Florida, USA, 2016

Standardized Provenance for Reproducible Dataflows in Neuroscience
Japan Neuroscience Society, Yokohama, Japan, 2016

Speaking one's mind: Vocal biomarkers of depression and Parkinson disease
Acoustical Society of America, Salt Lake City, USA, 2016

Predicting Treatment Outcome in Anxiety and Depression
McLean Hospital, Belmont, USA, 2015
Organization for Human Brain Mapping, Hawaii, USA, 2015

Linking Knowledge and Reproducible Research Via Standardized Provenance Models
Workshop at the Bernstein Computational Neuroscience conference, Heidelberg, Germany, 2015
Tools for Integrating and Planning Research in Neuroscience, UCLA, Los Angeles, USA, 2014

A Neuroinformatics Bridge to Personalized Healthcare
Boston University, Hearing research seminar, Boston, USA, 2014
Vanderbilt University, Nashville, USA, 2014

Enabling knowledge generation and reproducible research by embedding provenance models in metadata stores
Neuroinformatics Congress, Stochkholm, Sweden, 2013

Python Tools for Reproducible Research in Brain Imaging
PyData conference, Boston, USA, 2013

Nipype: Opensource platform for unified and replicable interaction with existing neuroimaging tools

Brigham and Womens Hospital, Boston, USA, 2009
Massachusetts General Hospital, Boston, USA, 2010, 2012, 2013
Radiology, U of Washington, Seattle, USA, 2011,
PICSL, U of Pennsylvania, Philadelphia, USA, 2011
Scientific Python Conference in India, Hyderabad, India, 2010
INCF Datasharing Workshop, Quebec, Canada, 2011
Python in Neuroscience Workshop, Paris, France, 2011

Leveraging scientific computation to bridge neuroimaging and clinical applications
Radiology, U of Pennsylvania, Philadelphia, USA, 2011
Haskins Laboratories, New Haven, Connecticut, USA 2012

Datasharing and reproducible research: Barriers and solutions
Janelia Farm Bioimage Informatics II Conference, Washington DC, USA, 2011
University de Montreal, Montreal, Canada, 2013

Using high-resolution fMRI to identify individual-specific speech motor regions
Surgical Brain-Mapping laboratory, Brigham and Womens Hospital, Boston, USA, 2010

Region of interest analysis of functional Magnetic Resonance Imaging data
New York State Psychiatric Institute, Columbia University, New York, USA, 2007
Singapore General Hospital, Singapore, Singapore, 2007

Exploring speech motor control through computational modeling and neuroimaging
Center for Life Sciences, National University of Singapore, Singapore, 2007

Research contracts and grants
Current

2025 - 2030 Center for Multi-Scale Multi-Omic Human Brain Atlas
NIH/NIMH/UM1 MH134812
Co-I

2025 – 2026 A Multi-Modal AI Approach to Measure Language and Social Communication Trajectories from Home Videos
Simons Foundation
Site PI
$125,000 (MIT Sub)

2025 - 2026 AI-Driven Tutors and Open Datasets for Early Literacy Education
MIT GenAI Consortium
$75,000

2024 - 2029 BBQS AI Resource and Data Coordinating Center (BARD.CC)
NIH/NIMH/U24 MH136628
Lead PI (MPI Laura Cabrera - Penn State University, David Kennedy - UMass Medical School)
$1,993,419 (2024 - 2025) Expected total: $9,984,860

2023 - 2028 BRAIN Connects: The center for Large-scale Imaging of Neural Circuits (LINC)
NIH/NINDS/UM1 NS132358
Site PI
$364,155 (MIT Sub 2023 - 2024) Expected Total MIT Sub (2028): $2,167,536

2022 – 2026 Voice as a Biomarker of Health: Building an ethically sourced, bioacoustic database to understand disease like never before
NIH/OD/OT2 OD032720
MPI (co-PIs: see https://reporter.nih.gov/search/FS74E19pCEacUekazZnwjA/project-details/10858564)
$9,119,334 (2022 – 2024; Budgets renegotiated each year)

2022 – 2027 An extensible brain knowledge base and toolset spanning modalities for multi-species data-driven cell types
NIH/NIMH/U24MH130918
MPI (co-PI Shoaib Mufti, Michael Hawrylycz, Lydia Ng, Allen Institute for Brain Science)
$6,146,053 (2022 – 2025) Expected Total (2027): $10,407,993

2022 – 2025 ReadNet: Preventing Reading Failure
Harvard University (Gift from Schmidt Futures and Citadel)
Site Co-Investigator
Total: $1,750,000

2019 – 2029 DANDI: Distributed Archives for Neurophysiology Data Integration
NIH/NIMH/R24 MH117295
Lead PI (MPI - Yaroslav Halchenko, Dartmouth College)
$8,309,010 (2019 – 2025) Expected Total (2029): $15,622,611

2016 – 2026 ReproNim: Center for Reproducible Neuroimaging Computation
NIH/NIBIB/P41 EB019936 (PI: David Kennedy, UMass Medical School)
Director: Technology, Research and Development Project 2
Site PI
$1,588,558 (MIT Sub 2016 - 2024)

Past

2008 – 2010 Dissemination of cross-platform software for artifact detection and region of interest analysis of fMRI data
NIH/NIBIB/R03 EB008673
PI (Co-PI Susan Whitfield-Gabrieli, McGovern Institute for Brain Research, MIT)
$162,238

2012 – 2014 Learned regulation of the limbic network via combined EEG and fMRI (PI: John Gabrieli)
NIH/NIMH/R21 MH092564
Co-Investigator
$431,745

2012 – 2015 MURFI: An Optimized Platform for Realtime fMRI Neurofeedback
MIT McGovern Institute Neurotechnology Program
Co-PI with John Gabrieli (MIT), Eden Evins (MGH)
$100,000

2011 – 2016 Using Real-Time Functional Brain Imaging and Computer Training To Enhance Recovery from Traumatic Brain Injury (TBI) (PI: John Gabrieli)
DOD/Clinical trial award PT100120
Co-Investigator
$2,800,000

2015 – 2017 Genetic Determinants of Schizophrenia Intermediate Phenotypes
NIH/NIMH/R01 (Supplement) MH092380 (PI: Tracey Petryshen, MGH)
Site PI
$170,167 (MIT Sub)

2012 – 2017	A randomized controlled trial of intranasal oxytocin as an adjunct to behavioral therapy for autism spectrum disorder (PI: John Gabrieli, MGH) DOD/Clinical Trial Award AR110329 Site PI $74,450 (MIT Sub)
2014 – 2017	Brain basis for voice-based tracking of neurological disorders MIT McGovern Institute Neurotechnology Program MIT Lincoln Lab Funds PI (Co-PI, Tom Quatieri, MIT Lincoln Laboratory, MIT) $150,000
2012 – 2018	Blast Induced Traumatic Brain Injury DOD/Institute for Soldier Nanotechnologies Investigator $450,000
2015 – 2020	Connectomes related to anxiety and depression in adolescents. NIH/NIMH/U01 MH108168 (PI: Susan Whitfield-Gabrieli, John Gabrieli, MIT) Informatics Lead $4,569,962
2016 – 2020	Nipype: Dataflows for Reproducible Biomedical Research NIH/NIBIB/R01 EB020740 PI $2,678,253
2019 – 2020	Tracking Alzheimer's Disease from Retinal OCT Images using Deep Learning Foundation for Ophthalmology Research and Education International, Inc. PI $30,000
2020 – 2021	Realtime speech modification apparatus for enhancing fluency in people that stutter. McGovern Institute Neurotechnology Program PI (co-PI Tod Machover, Media Arts and Sciences) $150,000
2019 - 2022	The Neuroimaging Data Model: FAIR descriptors of Brain Initiative Imaging Experiments (PI: David Keator, University of California, Irvine) NIH/NIMH/RF1 MH120021 Site PI $178,170 (MIT Sub), Total $943,951
2016 – 2022	NeuroScout: A cloud-based platform for rapid re-analysis of naturalistic fMRI datasets NIH/NIMH/R01 MH109682 (PI: Tal Yarkoni, UTexas, Austin) Site PI $431652 (MIT Sub), Total $3,026,872
2020 – 2023	Mumble Melody: A real-time speech modification system to enhance fluency in people who stutter MIT Deshpande Center for Technological Innovation PI (co-PI Tod Machover, Media Arts and Sciences) $213,064
2020 – 2024	Nobrainer: A robust and validated neural network tool suite for imagers NIH/NIMH/RF1 MH121885 PI $2,419,908

2023 – 2025 Biometric assessment and monitoring of psychiatric symptoms
Child Mind Institute, New York
PI
$299,348

Summary of Research
My research explores the intersection of computer science and neuroscience, with a focus on neuroinformatics, applied machine learning and signal processing, and translation of research to practical settings. My research addresses gaps in scientific knowledge and technologies in three areas:

1. **Neurophysiology and clinical applications of human communication:** This research investigates the neural underpinnings of spoken and written language and translates these insights into healthcare solutions. Key contributions include:
    - Development of the first neurocomputational speech motor learning and control model that accounted for the impact of transmission delays in the human brain and physiology.
    - Establishment of identification of individual-specific nuclei in the brainstem crucial for listening and spoken communication using high-resolution MRI.
    - Demonstration that specific brain regions modulate their activity in relation to symptom severity while speaking, providing a neural foundation for the relationship between speech and mood.
    - Development of an apparatus for transforming real time vocal feedback to alleviate stuttering, and received a US Patent for this technology.
2. **Machine learning and AI for personalized medicine:** This research aims to integrate multimodal data using machine learning and AI algorithms for clinical diagnosis, prognosis, and treatment or outcome prediction. Key contributions include:
    - Use of brain imaging, voice/speech recordings, and text analysis to demonstrate that we can track different aspects of health including brain tumors, depression, anxiety, autism, Parkinson's disease, dementia, suicidal ideation, and voice and respiratory disorders.
    - Development of new privacy preserving distributed machine learning algorithms and deep learning frameworks for neuroimaging that clinician scientists can use.
    - Demonstration that real-time fMRI biofeedback can assess and intervene individually to reduce hallucinations in schizophrenia and to reduce symptoms of depression in children.
    - Use of natural language processing to detect suicidal thoughts and behaviors in servicepersons through their interactions on a social media platform.
3. **Neuroinformatics and open science for preserving information and generating shared knowledge:** This research focuses on establishing community standards and data models for organizing scientific information, creating performant tools that address the data and information deluge, and enhancing reproducible research. Key contributions include:
    - Development of some of the most widely used community data standards (e.g., the Brain Imaging Data Standard), analysis software (e.g., Nipype), and research infrastructure.
    - Development and operation of DANDI, a state of the art infrastructure that is used as a computational data repository and for collaborative research in cellular neurophysiology.
    - Creation of a knowledge infrastructure for neuroscience to connect assertions or claims to their evidence, especially to underlying data.

The research aims to redefine how we assess, track, and treat mental health and neurological disorders using a combination of scalable, personal sensors, longitudinal data, and AI technologies inspired by neuroscientific insight and knowledge. The ultimate goal is to develop biomarker technologies to help clinicians worldwide better diagnose and track brain illness and improve outcomes of individuals with such disorders. We have done so by bringing the complex world of neuroscience and neurotechnologies together through efficient and interactive infrastructure, and collaborative projects.